MILITARY SPECIFICATION

INCREMENTS, PROPELLENT M5, T8 AND T9

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers propellent increments for use in mortar ammunition.

1.2 Classification - Propellent increments shall be of the following designations as specified (see 6.1):

M5 for use in 81-millimeter (mm.) Shell, M362 with M141 Fin.
T8 for use in 105-mm. Shell, T53 with T10E5 Fin.
T9 for use in 60-mm. Shell, M49A2 with T5 Fin.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Flat; Steel
QQ-S-790 - Strapping, Round; Steel, Bare and Zinc Coated

MILITARY

JAN-P-108 - Packaging and Packing for Overseas Shipment, Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-B-117 - Bags, Interior Packaging
MIL-B-2427 - Boxes, Ammunition Packing; Wood, Nailed
MIL-G-2550 - General Specification for Ammunition, Except Small Arms Ammunition
MIL-B-3106 - Board, Composition, Water-Resistant, Solid

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to Commander, US Army Armament, Research and Development Center, ATTN SMCAR-ESC-AS, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-129 - Marking of Shipments for Storage

DRAWINGS

ORDNANCE CORPS

71-12-28 - Increment, Propellent, M5, Assembly and Details
71-12-29 - Increment, Propellent, T9, for 60mm. Shell, M49A2 with T5 Fin Assembly
71-12-31 - Increment, T8; Assembly and Details
F8796522 - Marking Diagram and Sealing for Wood Packing Boxes

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 Sampling Tables and Procedures for Continuous Sampling by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids, shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commision regulations are now a part of the Code of Federal Regulations (1949 Edition - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Order for the above publication should cite "49CFR 71-78 (Rev. 1950)")

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3. REQUIREMENTS

3.1 Material - Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Increments - The increments shall comply with all requirements specified on Drawing (dwg.) 71-12-28, 71-12-29 or 71-12-31, as applicable, and with all requirements specified in applicable specifications except as modified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions - The detailed quality assurance provisions of this specification are applicable to inspection performed by or for the Government and shall be performed as prescribed below. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Contractor Quality Assurance System - The contractor shall provide and maintain an effective Quality Assurance System acceptable to the Government covering the supplies under this contract. Results of all examinations and tests performed under the Quality Assurance System shall be provided. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when as a minimum, it provides the Quality Assurance required by this specification and other documents referenced in this specification and also provides for 100 percent examination of product for each listed critical defect. The contractor shall not be restricted to the inspection station nor to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute, as to whether certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of and obtain approval for any changes to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.1.1 At the time the lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all Quality Assurance Provisions of the written plan.
b. Number of units of product inspected
c. Results obtained for all inspections performed
d. Drawing and specification numbers and dates, together with an identification and date of changes
e. Certificates of analysis
f. Number of items in lot
g. Date submitted

4.1.1.2 The certificate shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principle.

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4.1.2 Government Verification Inspection

4.1.2.1 At the beginning of the contract, Government verification will consist of: (a) all inspections listed herein which are not reserved for sole performance by the contractor, (b) a monitoring of those inspections reserved for sole performance by the contractor, and (c) monitoring of the contractor's inspection and quality control operations. Monitoring will be defined in 4.1.2.2.

4.1.2.2 After the contracting officer has determined that the conditions for utilization of reduced verification have been met, Government verification will consist of the following: Using the contractor's written Quality Assurance Procedure, this specification and other contractual documents as a guide, the Government inspector shall monitor all quality assurance operations performed by the contractor, at regular but unscheduled intervals. Monitoring consists of surveillance of the operations to determine whether the practices, methods, and procedures of the written procedure are being properly applied. Deviations from the prescribed or agreed upon procedures, or instances of poor practices which might have an effect upon the quality of the product shall be immediately called to the attention of the contractor. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the inspector will report such instances. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. As an adjunct to his monitoring activities the inspector may conduct inspection, at unscheduled, random intervals, on product from lots which have been inspected and found to be acceptable by the contractor.

4.1.2.3 Government verification of listed critical characteristics will at all times be performed as specified below:

4.1.2.3.1 Inspection will be performed by a sampling plan with an AQL of 0.015 percent. Any plan from Standard MIL-STD-105 or Publication ORD-M608-11 may be used. The screening inspection normally required of the contractor in conjunction with plans from ORD-M608-11 or in case of rejection by MIL-STD-105 plans will be in addition to the contractor inspection performed prior to presentation.

4.2 Inspection Provisions

4.2.1 Lot - A lot shall consist of increments manufactured under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain increments of one type only manufactured of propellant from not more than one lot.

4.2.2 Examination - Examination inspection will be performed as specified in MIL-G-2550 and as indicated below. Sampling criteria will be in accordance with Standard MIL-STD-105 or Publication ORD-M608-11 as appropriate.

4.2.2.1 Loading assembly (see dwg. 71-12-28, 71-12-29, or 71-12-31)

Categories and defects	Method of Inspection
Critical	
a. Charge weight outside of tolerance by more than 8 percent of prescribed weight . . .	Approved Weighing Device
b. Propellant exposed or escaping from bag . . .	Visual

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4.2.2.1 Loading assembly (see dwg 71-12-28, 71-12-29, or 71-12-31) (Cont)

	Categories and defects	Method of inspection
	AQL 0.25 percent	
Major		
1.	Charge weight outside of tolerance by 8 percent or less of prescribed weight . .	Approved weighing device
2.	Button hole missing (71-12-29 only) . . .	Visual
Minor	AQL 4.00 percent	
41.	Length between button holes (except 71-12-29).	Scale
42.	Length of button hole, min	Scale
43.	Location of button hole	Scale
44.	Location of button (71-12-29 only)	Scale
45.	Thread end length of cross seam incorrect.	Scale
46.	Stitches broken or missing	Visual
47.	Foreign matter	Visaul

4.2.2.2 Sealed inner liner (see 5.1.1, 5.1.2, or 5.1.3)

	Categories and defects	Method of inspection
	AQL 0.25 percent	
Major		
1.	Liner damaged so that contents are exposed . or liable to become exposed	Visual
2.	Bag incompletely sealed	Visual

4.2.2.3 Sealed wood packing box (see 5.1.1, 5.1.2, or 5.1.3)

	Categories and defects	Method of inspection
	AQL 0.40 percent	
Major		
1.	Strapping missing, broken or loose	Visual-Manual
2.	Board broken or split	Visual
3.	Contents exposed	Visual
Minor	AQL 1.50 percent	
41.	Strapping improperly engaged	Visual-Manual
42.	Contents move when shaken	Manual
43.	Marking missing, misleading, or illegible	Visual

4.3 Inspection Equipment

4.3.1 Government Verification Inspection - Government verification inspection will be accomplished with the use of gages and equipment that has been certified, by the contracting officer, to the design specified in the "List of Inspection Equipment, number _____."

4.3.2 Contractor inspection - Unless otherwise specified herein, contractor inspection will be accomplished with the use of gages and equipment designed to perform, to the required accuracy, the inspection and controls delineated in the contractor's written description specified in 4.1.1. In inspecting the contractor's inspection equipment the Government will determine that the contractor has available and utilized correctly,

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4.3.2 (Cont)
gaging, measuring and test equipment of required accuracy and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Ordnance Personnel.

4.4 Tests Procedures - There are no tests procedures applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.1)

5.1.1 Level A - military pack - The propellent increments shall be packed in grade A, type 1, style 4 wooden packing boxes conforming to Specification MIL-B-2427 except that the bottom, sides and top shall be ½ inch lumber. The ends and cleates shall be 25/32 inch lumber. Two ½ inch thick by 1-3/4 inch wide cleates shall run across the cover located approximately 1-1/2 icnhes from each end. Each wooden box shall be provided with an inner bag of flexible water-vaporproof barrier material conforming to type I, class e of Specification MIL-B-117, which is heat sealed after filling. The length and width of the inner liner shall be approximately 1/4 inch greater than the inside length and width of the box. The depth of the inner liner shall be such that after heat sealing, sufficient looseness will remain to insure that the contents will not apply pressure to the inner liner or seams. Both ends and top of the wooden box shall be lined, between box and inner liner, with 1/16 inch thick composition board conforming to Specification MIL-B-3106. The wooden box shall be strapped girthwise, inside cover cleats, with two 5/8 inch by .02 inch thick steel straps type I, class B, conforming to Specification QQ-S-781 or alternatively, round zinc coated steel strapping number 14 (.08) gage, class B, conforming to Specification QQ-S-790 may be used. The gross weight of the box plus propellent shall not exceed 50 pounds.

5.1.2 Level B - Limited military pack - The propellent increments shall be packed in an approved stainless steel lined propellent container. The M25 propellant container is approved for this use. Alternatively the M18 propellant container may be used.

5.1.3 Level C - Minimum Military Pack - The increments shall be packed as specified in 5.1.2.

5.2 Marking - Level A packs shall be marked in accordance with Dwg. F8796522. ICC nomenclature shall be, "Propellant Explosives - Class A." Level B and Level C packs shall be marked in accordance with the requirements for the individual containers. ICC nomenclature is the same as for Level A packs.

5.3 Data cards - Data card information shall be as specified in Specification MIL-G-2550.

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6. NOTES

6.1 Ordering data - Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type required (see 1.2).
c. Levels of protection (see 5.1)

Notice - This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance, Department of the Army.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-45170	INCREMENTS, PROPELLANT M5, T8 AND T9

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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